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Inventor(s)	Patel; Vinesh Premji et al.

System and method for digitally marking artificial intelligence (AI) generated content

Abstract

A processor receives request to generate content using an AI tool and generates the requested content using the AI tool. The processor detects that the AI tool used a first AI module and a second AI module associated with the AI tool to generate the requested content. In response, the processor accesses, from a memory, a first digital signature associated with the first AI module and a second digital signature associated with the second AI module. The processor generates a combined digital signature based on the first digital signature and the second digital signature and embeds the combined digital signature in the generated content to generate a marked content.

Inventors: Patel; Vinesh Premji (Greenwich, GB), Young; Michael R (Davidson, NC)

Applicant: Bank of America Corporation (Charlotte, NC)

Family ID: 95821874

Assignee: Bank of America Corporation (Charlotte, NC)

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to data processing, and more specifically to a system and method for digitally marking artificial intelligence (AI) generated content.

BACKGROUND

(2) As the technology behind Artificial Intelligence (AI) continues to advance, so does the potential of its misuse. Generative AI is a subset of AI that creates new data or content such as images, audio files, digital art, and software code based on user prompts. This form of AI is already being used to generate fake and plagiarized content. For example, generative AI technology has been used to create deepfakes to spread disinformation. Further fake news and plagiarized digital art has also been created and spread across social media platforms. This has led to serious concerns about the authenticity of digital content and its impact on the public at large. With AI technology being constantly refined, these problems will only get worse with time.

SUMMARY

(3) The system and method implemented by the system as disclosed in the present disclosure provide technical solutions to the technical problems discussed above by identifying digital content generating using AI technology.

(4) For example, the disclosed system and methods provide the practical application of intelligently identifying a source of a digital content generated using AI technology. As described in further detail below, these techniques include digitally marking AI generated content to identify a source of the content. Digitally marking AI generated content includes intelligently adding a digital signature to the AI generated content, wherein the digital signature may be used to verify a known source of the content or to identify an unknown source of the content. Additionally, the disclosed techniques may be used to determine whether a piece of digital content was generated using an AI tool. In other words, the disclosed techniques may be used to determine whether a piece of digital content was generated using AI technology.

(5) As described in embodiments of the present disclosure, an AI tool may be configured to digitally mark AI content generated by the AI tool. In this context, the AI tool may store a primary digital signature associated with the AI tool that is unique to the AI tool. The primary digital signature may include and/or indicate information that identifies a source associated with the AI content. AI tool may be configured to embed the primary digital signature in the AI content. When the primary digital signature is decoded by a software application configured to consume digital content, the information decoded from the primary digital signature may allow an identification of the source of the AI content. For example, the information decoded from the primary digital signature may include a unique identity of the AI tool used to generate the AI content. This allows a determination that the particular piece of digital content was generated using AI technology (e.g., using an AI tool). Additionally, the information allows an identification of the particular AI tool used to generate the AI content. In an additional or alternative embodiment, the information decoded from the primary digital signature may include a unique identity of the computing node that generated the AI content and/or an identity of the owner of the AI content. The identification of these aspects associated with a piece of digital content avoids and/or prevents deep fakes, plagiarism, and fake information and thus improves data security on the internet.

(6) The disclosed system and methods provide the practical application of identifying AI modules used to generate an AI content. As described in embodiments of the present disclosure, the AI tool generates a combined digital signature including individual module signatures associated with each AI module used to generate the AI content. Inclusion of the module signatures allows identification of the particular AI modules that are used to generate AI content. This information may be used to re-train and refine the AI tool and improve generation of AI content. For example, a decoding algorithm running at a computing node may decode the combined digital signature embedded in the AI content, extract the individual module signatures, and identify the AI modules that were used by the AI tool to generate the AI content. This data may then be fed as training data to the AI tool to refine generation AI content. Improving generation of AI content improves the efficiency of the AI tool and the computing node running the AI tool.

(7) Thus, the disclosed system and method generally improves Artificial Intelligence technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 is a schematic diagram of a system, in accordance with certain aspects of the present disclosure; and

(3) FIG. 2 illustrates a flowchart of an example method for digitally marking AI content, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(4) FIG. 1 is a schematic diagram of a system **100**, in accordance with certain aspects of the present disclosure. As shown, system **100** includes a computing infrastructure **102** connected to a network **180**. Computing infrastructure **102** may include a plurality of hardware and software components. The hardware components may include, but are not limited to, computing nodes **104** such as desktop computers, smartphones, tablet computers, laptop computers, servers and data centers, mainframe computers, virtual reality (VR) headsets, augmented reality (AR) glasses and other hardware devices such as printers, routers, hubs, switches, and memory all connected to the network **180**. Software components may include software applications that are run by one or more of the computing nodes **104** including, but not limited to, operating systems, user interface applications, third party software, database management software, service management software, mainframe software, metaverse software, AI tools (e.g., AI tool **108**) and other customized software programs implementing particular functionalities. For example, software code relating to one or more software applications may be stored in a memory device and one or more processors (e.g., belonging to one or more computing nodes **104**) may execute the software code to implement respective functionalities. For example, software applications run by one or more computing nodes **104** of the computing infrastructure **102** may include Artificial Intelligence (AI) tools **108** and decoding algorithms **110**. In one embodiment, at least a portion of the computing infrastructure **102** may be representative of an Information Technology (IT) infrastructure of an organization.

(5) One or more of the computing nodes **104** may be operated by a user **106**. For example, a computing node **104** may provide a user interface using which a user **106** may operate the computing node **104** to perform data interactions within the computing infrastructure **102**.

(6) One or more computing nodes **104** of the computing infrastructure **102** may be representative of a computing system which hosts software applications that may be installed and run locally or may be used to access software applications running on a server (not shown). The computing system may include mobile computing systems including smart phones, tablet computers, laptop computers, or any other mobile computing devices or systems capable of running software applications and communicating with other devices. The computing system may also include non-mobile computing devices such as desktop computers or other non-mobile computing devices capable of running software applications and communicating with other devices. In certain embodiments, one or more of the computing nodes **104** may be representative of a server running one or more software applications to implement respective functionality (e.g., AI tool **108**) as described below. In certain embodiments, one or more of the computing nodes **104** may run a thin client software application where the processing is directed by the thin client but largely performed by a central entity such as a server (not shown).

(7) Network **180**, in general, may be a wide area network (WAN), a personal area network (PAN), a cellular network, or any other technology that allows devices to communicate electronically with other devices. In one or more embodiments, network **180** may be the Internet.

(8) At least a portion of the computing infrastructure **102** may include a blockchain network **190**.

For example, a portion of the computing nodes **104** may form the blockchain network **190**. As shown in FIG. **1**, example blockchain network **190** includes computing nodes **104a**, **104b**, **104c**, **104d**, **104e** and **104f** connected to each other via a portion of the network **180** (shown as **180a**). The blockchain network **190** implements distributed computing which generally refers to a method of making multiple computers (e.g., computing nodes **104a-104f**) work together to solve a common problem. This makes a computer network (e.g., blockchain network **190**) appear as a powerful single computer that provides large-scale resources to deal with complex challenges. For example, distributed computing can encrypt large volumes of data, solve complex physics and chemical equations with many variables, and render high-quality, three-dimensional video animation. Distributed computing often uses specialized software applications that are configured to run on several computing nodes **104** instead of on just one computer, such that different computers perform different tasks and communicate to develop the final solution. High-performing distributed computing is often used in engineering research, financial services, energy sector and the like to run complex processes.

(9) Embodiments of the present disclosure describe techniques for intelligently identifying a source of a digital content generated using AI technology. As described in further detail below, these techniques include digitally marking AI generated content to identify a source of the content. Digitally marking AI generated content includes intelligently adding a digital signature to the AI generated content, wherein the digital signature may be used to verify a known source of the content or to identify an unknown source of the content. Additionally, the disclosed techniques may be used to determine whether a piece of digital content was generated using an AI tool **108**. In other words, the disclosed techniques may be used to determine whether a piece of digital content was generated using AI technology.

(10) At least a portion of the computing infrastructure **102** (e.g., one or more computing nodes **104**) may implement an AI tool **108** which may perform a plurality of operations associated with generating and digitally marking AI content **124**. The AI tool comprises a processor **162**, a memory **166**, and a network interface **164**. The AI tool **108** may be configured as shown in FIG. **1** or in any other suitable configuration.

(11) The processor **162** comprises one or more processors operably coupled to the memory **166**. The processor **162** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **162** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **162** is communicatively coupled to and in signal communication with the memory **166**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **162** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **162** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(12) The one or more processors are configured to implement various instructions, such as software instructions. For example, the one or more processors are configured to execute instructions (e.g., AI tool instructions **146**) to implement the AI tool **108**. In this way, processor **162** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the AI tool **108** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The AI tool **108** is configured to operate as described with reference to FIG. **2**. For example, the processor **162** may be configured to perform at least a portion of the method **200** as described in FIG. **2**.

(13) The memory **166** comprises a non-transitory computer-readable medium such as one or more

disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **166** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(14) The memory **166** is operable to store primary digital signature **122**, AI content **124**, AI modules **130**, combined digital signature **142**, hashing algorithm **144**, and the AI tool instructions **146**. The AI tool instructions **146** may include any suitable set of instructions, logic, rules, or code operable to execute the AI tool **108**.

(15) The network interface **164** is configured to enable wired and/or wireless communications. The network interface **164** is configured to communicate data between the AI tool **108** and other devices, systems, or domains (e.g., computing nodes **104**). For example, the network interface **164** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **162** is configured to send and receive data using the network interface **164**. The network interface **164** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(16) It may be noted that each of the computing nodes **104** may be implemented like the AI tool **108** shown in FIG. 1. For example, each of the computing nodes **104** may have a respective processor and a memory that stores data and instructions to perform a respective functionality of the computing node **104**.

(17) AI tool **108** may be configured to generate digital content (e.g., AI content **124**) using AI technology (e.g., machine learning) based on user prompts (e.g., prompts made by a user **106**) or machine generated prompts. Example, prompts made to the AI tool **108** may include, but are not limited to, a text prompt and a voice prompt. The user prompts and machine generated prompts are collectively shown in FIG. 1 as request **112**. Upon receiving a request **112** (e.g., a text prompt), the AI tool **108** may be configured to generate AI content **124** that may include but is not limited to, pictures, text (e.g., text document **124a** and/or software code), audio clips, and video clips. For example, a user **106** may enter a text prompt on a user interface provided by the AI tool **108**, wherein the user prompt may ask the AI tool **108** to generate a picture of a lion walking in a rainforest. The AI tool **108** may generate a computer-generated digital picture in response to receiving the text prompt.

(18) In certain embodiments, AI tool **108** may include a plurality of AI modules **130** (shown as including first AI module **130a**, second AI module **130b**, . . . Nth AI module **130n**), wherein each AI module **130** is configured (e.g., trained) to generate a particular type of AI content **124** or a portion thereof. In other words, each AI module **130** is trained to generate a particular type of AI content **124** including, but not limited to, pictures, text, audio, video, graphs, or tables. For example, the first AI module **130a** may be trained to generate pictures, and the second AI module **130b** may be trained to generate text. Similarly, other AI modules **130** of the AI tool **108** may be trained to generate audio, video, graphs, and tables. The AI tool **108** may use multiple AI modules **130** to generate a requested AI content **124**. For example, when a user **106** prompts the AI tool **108** to generate a scientific report based on provided input data, the AI tool **108** may use a first AI module to generate text, a second AI module to generate pictures, a third AI module to generate tables, and a fourth AI module to generate graphs/plots. The AI tool **108** may then put together the individual portions generated by the individual module to generate the requested report.

(19) AI tool **108** may be configured to digitally mark AI content **124** generated by the AI tool **108**. In this context, AI tool **108** may store a primary digital signature **122** associated with the AI tool **108** and that is unique to the AI tool **108**. The primary digital signature **122** may include and/or indicate information that identifies a source associated with the AI content **124**. For example, the primary digital signature **122** may include one or more of a unique identity of the AI tool **108** (e.g.,

tool name, version etc.), an identity of the computing node **104** that was used to generate the AI content **124**, or identity of an owner of the AI content **124** (e.g., copyright information). It may be noted that this is not an exhaustive list and that the primary digital signature **122** may include or indicate a multitude of other information as may be envisaged by a person of ordinary skill in the art.

(20) AI tool **108** may be configured to embed the primary digital signature **122** in the AI content **124**. Embedding the primary digital signature **122** in the AI content **124** may include one or more of adding a digital watermark including the primary digital signature **122** to the AI content **124**, adding a digital fingerprint including the primary digital signature **122** to the AI content **124**, or adding a steganographic code including the primary digital signature **122** to the AI content **124**. A person having ordinary skill in the art may appreciate that other known methods for embedding data into digital content may be used to embed the primary digital signature **122** into the AI content **124**. Once digitally marked, the marked AI content **124** may be transmitted to any other computing node **104** of the computing infrastructure **102**.

(21) One or more computing nodes **104** of the computing infrastructure **102** may be configured to open and review the AI content **124**. For example, a software application may be installed at a particular computing node **104** that is configured to open and review the AI content **124**.

Depending on the nature of the AI content **124** (e.g., text, pictures, audio, video etc.), the software application may be a digital media player, a text editor, word processor, a pdf reader and/or a multi-media player. The software application may include a decoding algorithm **110** that is configured to decode digital signatures (e.g., primary digital signature **122**). For example, the decoding algorithm **110** may be included as a plug-in software to the software application that is to be used to review the AI content **124**. The decoding algorithm **110** may be configured to automatically run when any content is opened by the software application including AI content **124**. For example, in response to receiving a command to open the marked AI content **124**, the software application may open the AI content **124**. This may automatically trigger the decoding algorithm **110** to extract and decode the digital signature embedded in the AI content **124**. For example, when a digital signature (e.g., primary digital signature **122**) is included (e.g., embedded) in a piece of digital content (e.g., AI content **124**) being opened by a software application, the decoding algorithm **110** automatically detects and decodes the digital signature to extract information included in the digital signature. For example, the decoding algorithm **110** is configured to detect and decode the primary digital signature **122** that is embedded in the digitally marked AI content **124**. Based on decoding the primary digital signature **122**, the decoding algorithm **110** extracts the information included in the primary digital signature **122**.

(22) The information decoded from the primary digital signature **122** may allow an identification of the source of the AI content **124**. For example, the information decoded from the primary digital signature **122** may include a unique identity of the AI tool used to generate the AI content **124**. This allows a determination that the particular piece of digital content (e.g., AI content **124**) was generated using AI technology (e.g., using an AI tool **108**). Additionally, the information allows an identification of the particular AI tool **108** used to generate the AI content **124**. In an additional or alternative embodiment, the information decoded from the primary digital signature **122** may include a unique identity of the computing node **104** that generated the AI content **124** and/or an identity of the owner of the AI content **124**. The identification of these aspects associated with a piece of digital content avoids and/or prevents deep fakes, plagiarism, and fake information.

(23) In certain embodiments, the primary digital signature **122** is changed from time to time (e.g., periodically or based on a pre-configured schedule) to prevent bad actors from acquiring the primary digital signature **122**, for example, by reverse engineering AI content **124** generated by the AI tool **108** and using it to generate fake AI content **124**.

(24) In one or more embodiments, the AI tool **108** may store a module signature **132** associated with each AI module **130**, wherein each module signature **132** is a digital signature that uniquely

identifies a particular AI module **130**. For example, a module signature **132** associated with a particular AI module **130** includes a unique identifier (e.g., module name, version etc.) of the AI module **130**. As shown in FIG. 1, the first AI module **130a** is associated with a first module signature **132a**, the second AI module **130b** is associated with a second module signature **132b**, and the Nth AI module **130n** is associated with Nth module signature **132n**. AI tool **108** may be configured to embed in the AI content **124**, the module signature **132** of each AI module **130** that is used to generate the AI content **124**. In one embodiment, AI tool **108** generates a combined digital signature **142** based on the module signatures **132** associated with AI modules **130** used to generate the AI content **124** and embeds the combined digital signature **142** in the AI content **124**. For example, when the first AI module **130a** and the second AI module **130b** are used to generate the AI content **124**, AI tool **108** generates a combined digital signature **142** based on the first module signature **132a** and the second module signature **132b**. In one embodiment, the combined digital signature **142** includes the module signatures **132** associated with the AI modules **130** used to generate the AI content **124**. Inclusion of the module signatures **132** allows identification of the particular AI modules **130** that are used to generate AI content **124** based on a particular prompt (e.g., text prompt). This information may be used to re-train and refine the AI tool **108** and improve generation of AI content **124**. For example, a decoding algorithm **110** running at a computing node **104** may decode the combined digital signature **142** embedded in the AI content **124**, extract the individual module signatures **132**, and identify the AI modules **130** that were used by the AI tool **108** to generate the AI content **124**. This data may then be fed as training data to the AI tool **108** to refine generation AI content **124**. In an additional or alternative embodiment, the inclusion of the module signatures **132** in the AI content **124** indicates that the AI content **124** was generated using AI technology (e.g., using an AI tool **108**).

(25) In certain embodiments, the primary digital signature **122** may include a hash code **145** that is uniquely generated for a particular AI content **124**. For example, AI tool **108** may be configured to store and deploy a hashing algorithm **144** that generates a hashing code **145** based on pre-selected portions of a text document generated by the AI tool **108**. For example, the hashing algorithm **144** may be configured to generate a hash code **145** based on text at line 3 between spaces 2 and 5, line 8 between spaces 1 and 6, line 12 between spaces 4 and 11 etc. When a text document (e.g., software code) is generated using the AI tool **108**, the AI tool **108** generates a hash code **145** based on the hashing algorithm **144** and embeds the hash code **145** in the text document. Since every text document will have different text at the pre-selected portions, the same hashing algorithm **144** generates a unique hash code **145** for each text document, making it very difficult for a bad actor to identify the hashing algorithm **144** by reverse-engineering hash codes **145** embedded in text documents generated by the AI tool **108**. One example of a hashing algorithm **144** is a Secure Hashing Algorithm (SHA) that is generally used for hashing data.

(26) A software application (e.g., software code editor) that can be used to open and review text documents such as software code may include a software plug-in configured with the same hashing algorithm **144**. For example, a decoding algorithm **110** running at a computing node **104** may include the same hashing algorithm **144** that is used by the AI tool **108** to generate hash codes **145**. When the software application is used to open the text document, the decoding algorithm **110** reads the hash code **145** embedded in the text document and verifies the hash code **145** based on the text document. For example, the decoding algorithm **110** generates a second hash code based on the same hashing algorithm used by the AI tool **108** and compares the generated second hash code with the hash code **145** embedded in the text document. When the generated second hash code matches with the hash code **145** embedded in the text document, the decoding algorithm **110** determines that the hash code **145** has been verified and identifies the text document as generated by the particular AI tool **108**.

(27) The AI tool **108** may be configured to periodically change the hashing algorithm **144** that is used to generate the hashing code **145**. This avoids a potential attacker/hacker from identifying the

hashing algorithm by reverse engineering the hash codes **145** embedded in text documents generated by the AI tool **108**. For example, from time to time, the AI tool **108** may change the pre-selected portions (e.g., lines, word positions etc.) from AI generated text documents that is to be used to generate the hash code **145**. Each time the hashing algorithm **144** is changed by the AI tool **108**, the changed hashing algorithm **144** is transmitted to software applications that are configured to consume the AI content **124** generated by the AI tool **108**, so that the hash codes **145** can be verified using the latest hashing algorithm **144**. In some embodiments, the AI tool **108** may include (e.g., as part of the primary digital signature **122**) a date on which a particular piece of AI content **124** was generated. This allows a decoding algorithm **110** at the receiving end to use the correct hashing algorithm **144** that was used by the AI tool **108** to generate the AI content **124**, for verifying the AI content **124**. For example, a computing node **104** that runs a software application configured to consume AI content **124** may store several hashing algorithms **144** and the dates (e.g., date ranges) each hashing algorithm **144** was used by the AI tool **108** to generate hash codes **145**. Upon receiving a particular piece of AI content **124** (e.g., text document) the decoding algorithm **110** decodes the primary digital signature **122** and reads the hash code **145** along with the particular date on which the AI content **124** was generated. Then the decoding algorithm **110** looks up the hashing algorithm **144** that was used on that particular date and uses the hashing algorithm **144** to verify the received hash code **145**.

(28) In some cases, a bad actor may modify the text of a text document generated by the AI tool **108** in a non-substantive manner to avoid identification as being generated by the AI tool **108**. For example, when the text in a text document generated by the AI tool **108** is modified after the hash code **145** has been embedded in the text document, a second hash code generated by the decoding algorithm **110** of a consuming software application based on the modified text document may not match with the hash code **145** embedded in the text document. In this context, a software application configured to consume the AI content **124** generated by the AI tool **108** may be configured to perform an entropy analysis of the hash code **145** embedded in the text document and generate a similarity value that indicates how similar a second hash code generated at the receiving end is to the hash code **145** embedded in the text document, by comparing the second hash code generated at the receiving end to the hash code **145** embedded in the text document. When a similarity value is above a pre-configured threshold (e.g., 85%), it is determined that the text document was generated by the AI tool **108** and modified later. Thus, the entropy analysis allows identification of text documents generated by the AI tool **108** even when the text documents were modified from their original generated form. This avoids plagiarizing of AI content **124** generated by the AI tool **108** by bad actors by modifying the original AI content **124**.

(29) In certain embodiments, the AI content **124** generated by the AI tool **108** may be stored in the blockchain network **190**. Since data stored in a blockchain network **190** is generally immutable, this provides an additional layer of data security and avoids bad actors from faking or modifying AI generated content **124**.

(30) FIG. 2 illustrates a flowchart of an example method **200** for digitally marking AI content **124**, in accordance with one or more embodiments of the present disclosure. Method **200** may be performed by the AI tool **108** shown in FIG. 1.

(31) At operation **202**, AI tool **108** receives a request to generate AI content **124**.

(32) At operation **204**, the AI tool **108** generates the requested AI content **124**.

(33) As described above, AI tool **108** may be configured to generate digital content (e.g., AI content **124**) using AI technology (e.g., machine learning) based on user prompts (e.g., prompts made by a user **106**) or machine generated prompts. Example, prompts made to the AI tool **108** may include, but are not limited to, a text prompts and a voice prompts. The user prompts and machine generated prompts are collectively shown in FIG. 1 as request **112**. Upon receiving a request **112** (e.g., a text prompt), the AI tool **108** may be configured to generate AI content **124** that may include but is not limited to, pictures, text (e.g., text document **124a** and/or software code), audio clips, and video

clips. For example, a user **106** may enter a text prompt on a user interface provided by the AI tool **108**, wherein the user prompt may ask the AI tool **108** to generate a picture of a lion walking in a rainforest. The AI tool **108** may generate a computer-generated digital picture in response to receiving the text prompt.

(34) At operation **208**, the AI tool **108** detects that a first AI module **130a** and a second AI module **130b** associated with the AI tool **108** were used to generate the requested AI content **124**.

(35) In certain embodiments, AI tool **108** may include a plurality of AI modules **130** (shown as including first AI module **130a**, second AI module **130b**, . . . Nth AI module **130n**), wherein each AI module **130** is configured (e.g., trained) to generate a particular type of AI content **124** or a portion thereof. In other words, each AI module **130** is trained to generate a particular type of AI content including, but is not limited to, pictures, text, audio, video, graphs, or tables. For example, the first AI module **130a** may be trained to generate pictures, and the second AI module **130b** may be trained to generate text. Similarly, other AI modules of the AI tool **108** may be trained to generate audio, video, graphs, and tables. The AI tool **108** may use multiple AI modules **130** to generate a requested AI content **124**. For example, when a user **106** prompts the AI tool **108** to generate a scientific report based on provided input data, the AI tool **108** may use a first AI module to generate text, a second AI module to generate pictures, a third AI module to generate tables, and a fourth AI module to generate graphs/plots. The AI tool **108** may then put together the individual portions generated by the individual module to generate the requested report.

(36) At operation **208**, AI tool **108** accesses, from the memory **166** a first digital signature (e.g., first module signature **132a**) associated with the first AI module **130a** and a second digital signature (e.g., second module signature **132b**) associated with the second AI module **130b**.

(37) At operation **210**, AI tool **108** generates a combined digital signature **142** based on the first digital signature (e.g., first module signature **132a**) and the second digital signature (e.g., second module signature **132b**).

(38) At operation **212**, AI tool **108** embeds the combined digital signature **142** in the generated AI content **124** to generate a marked AI content.

(39) As described above, the AI tool **108** may store a module signature **132** associated with each AI module **130**, wherein each module signature **132** is a digital signature that uniquely identifies a particular AI module **130**. For example, a module signature **132** associated with a particular AI module **130** includes a unique identifier (e.g., module name, version etc.) of the AI module **130**. As shown in FIG. **1**, the first AI module **130a** is associated with a first module signature **132a**, the second AI module **130b** is associated with a second module signature **132b**, and the Nth AI module **130n** is associated with Nth module signature **132n**. AI tool **108** may be configured to embed in the AI content **124**, the module signature **132** of each AI module **130** that is used to generate the AI content **124**. In one embodiment, AI tool **108** generates a combined digital signature **142** based on the module signatures **132** associated with AI modules **130** used to generate the AI content **124** and embeds the combined digital signature **142** in the AI content **124**. For example, when the first AI module **130a** and the second AI module **130b** are used to generate the AI content **124**, AI tool **108** generates a combined digital signature **142** based on the first module signature **132a** and the second module signature **132b**. In one embodiment, the combined digital signature **142** includes the module signatures **132** associated with the AI modules **130** used to generate the AI content **124**. Inclusion of the module signatures **132** allows identification of the particular AI modules **130** that are used to generate AI content **124** based on a particular prompt (e.g., text prompt). This information may be used to re-train and refine the AI tool **108** and improve generation of AI content **124**. For example, a decoding algorithm **110** running at a computing node **104** may decode the combined digital signature **142** embedded in the AI content **124**, extract the individual module signatures **132**, and identify the AI modules **130** that were used by the AI tool **108** to generate the AI content **124**. This data may then be fed as training data to the AI tool **108** to refine generation AI content **124**. In one embodiment, the inclusion of the module signatures **132** in the AI content **124**

indicates that the AI content **124** was generated using AI technology (e.g., using an AI tool **108**).

(40) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(41) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(42) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a memory storing at least one digital signature associated with an Artificial Intelligence (AI) tool; and a processor communicatively coupled to the memory and configured to: receive a request to generate content using the AI tool, wherein the content comprises one or more of a picture, an audio clip, a video clip, a text document or software code; generate the requested content using the AI tool, wherein: the AI tool comprises a plurality of AI modules; each AI module is configured to generate a particular type of content; and each AI module is associated with a unique digital signature that uniquely identifies the AI module; detect that the AI tool used a first AI module and a second AI module associated with the AI tool to generate the requested content; access, from the memory, a first digital signature associated with the first AI module and a second digital signature associated with the second AI module; generate a combined digital signature based on the first digital signature and the second digital signature; and embed the combined digital signature in the generated content to generate a marked content.

2. The system of claim 1, wherein the processor is further configured to: access, from the memory, a primary digital signature associated with the AI tool, wherein the primary digital signature uniquely identifies the AI tool; and embed the primary digital signature in the marked content.

3. The system of claim 2, wherein the primary digital signature comprises or indicates information relating to one or more of an owner of the marked content and an identity of the AI tool.

4. The system of claim 1, wherein a digital signature associated with a particular AI module comprises or indicates an identity of the AI module.

5. The system of claim 1, wherein the processor is configured to embed the combined digital signature in the generated content by: one or more of adding a digital watermark to the content, adding a digital fingerprint to the content, or adding a steganographic code to the content.

6. The system of claim 1, further comprising a second processor configured to: receive a command to open the marked content; in response to receiving the command, extract the combined digital signature from the marked content; decode the combined digital signature to extract the first digital signature and the second digital signature; and based on decoding the combined digital signature: determine that the content was generated using an AI tool; and identify based on the first digital signature and the second digital signature that the AI tool used the first AI module and the second

AI module to generate the marked content.

7. The system of claim 1, further comprising a blockchain network, wherein the processor is further configured to store the marked content as a data interaction in a blockchain network.

8. The system of claim 1, wherein: the requested content is a text document; the memory is further configured to store a coding algorithm for generating hash codes based on text documents generated by the AI tool, wherein the coding algorithm is unique to the AI tool; the processor is further configured to: access the coding algorithm from the memory, wherein the coding algorithm is configured to generate a hash code based on pre-configured portions of the text document; generate the hash code based on the pre-configured portions of the text document using the coding algorithm; and embed the hash code in the text document to generate a marked text document.

9. The system of claim 8, wherein the processor is further configured to modify the coding algorithm according to a pre-configured schedule.

10. A method for digitally marking AI content, comprising: receiving a request to generate content using an Artificial Intelligence (AI) tool, wherein the content comprises one or more of a picture, an audio clip, a video clip, a text document or software code; generating the requested content using the AI tool, wherein: the AI tool comprises a plurality of AI modules; each AI module is configured to generate a particular type of content; and each AI module is associated with a unique digital signature that uniquely identifies the AI module; detecting that the AI tool used a first AI module and a second AI module associated with the AI tool to generate the requested content; accessing, from a memory, a first digital signature associated with the first AI module and a second digital signature associated with the second AI module; generating a combined digital signature based on the first digital signature and the second digital signature; and embedding the combined digital signature in the generated content to generate a marked content.

11. The method of claim 10, further comprising: accessing, from the memory, a primary digital signature associated with the AI tool, wherein the primary digital signature uniquely identifies the AI tool; and embedding the primary digital signature in the marked content.

12. The method of claim 11, wherein the primary digital signature comprises or indicates information relating to one or more of an owner of the marked content and an identity of the AI tool.

13. The method of claim 10, wherein a digital signature associated with a particular AI module comprises or indicates an identity of the AI module.

14. The method of claim 10, wherein embedding the combined digital signature in the generated content comprises one or more of adding a digital watermark to the content, adding a digital fingerprint to the content, or adding a steganographic code to the content.

15. The method of claim 10, further comprising: receive a command to open the marked content; in response to receiving the command, extract the combined digital signature from the marked content; decode the combined digital signature to extract the first digital signature and the second digital signature; and based on decoding the combined digital signature: determine that the content was generated using an AI tool; and identify based on the first digital signature and the second digital signature that the AI tool used the first AI module and the second AI module to generate the marked content.

16. A non-transitory computer-readable medium storing instructions that when executed by a processor cause the processor to: receive a request to generate content using an Artificial Intelligence (AI) tool, wherein the content comprises one or more of a picture, an audio clip, a video clip, a text document or software code; generate the requested content using the AI tool, wherein: the AI tool comprises a plurality of AI modules; each AI module is configured to generate a particular type of content; and each AI module is associated with a unique digital signature that uniquely identifies the AI module; detect that the AI tool used a first AI module and a second AI module associated with the AI tool to generate the requested content; access, from a memory, a first digital signature associated with the first AI module and a second digital signature associated with

the second AI module; generate a combined digital signature based on the first digital signature and the second digital signature; and embed the combined digital signature in the generated content to generate a marked content.

17. The non-transitory computer-readable medium of claim 16, wherein the instructions further cause the processor to: access, from the memory, a primary digital signature associated with the AI tool, wherein the primary digital signature uniquely identifies the AI tool; and embed the primary digital signature in the marked content.

18. The non-transitory computer-readable medium of claim 17, wherein the primary digital signature comprises or indicates information relating to one or more of an owner of the marked content and an identity of the AI tool.

19. The non-transitory computer-readable medium of claim 16, wherein a digital signature associated with a particular AI module comprises or indicates an identity of the AI module.

20. The non-transitory computer-readable medium of claim 16, wherein embedding the combined digital signature in the generated content comprises one or more of adding a digital watermark to the content, adding a digital fingerprint to the content, or adding a steganographic code to the content.
